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*
*          STAAD.Pro
*          Version 2007 Build 04
*          Proprietary Program of
*          Research Engineers, Intl.
*          Date= MAR 25, 2013
*          Time= 19:32:48
*
*          USER ID: Parsons Brinckerhoff
*****

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1. STAAD SPACE
INPUT FILE: 130222 Crane Staircase R2.STD
2. START JOB INFORMATION
3. ENGINEER DATE 26-NOV-12
4. JOB NAME CRANE STAIRCASE
5. END JOB INFORMATION
6. INPUT WIDTH 79
7. UNIT METER KN
8. JOINT COORDINATES
9. 1 -109.835 0 -23.25; 2 -109.835 0 -22.05; 3 -107.085 0 -23.25
10. 4 -107.085 0 -22.05; 5 -107.085 3 -23.25; 6 -107.085 3 -22.05
11. 7 -106.085 0 -23.25; 8 -106.085 0 -22.05; 9 -106.085 3 -23.25
12. 10 -106.085 3 -22.05; 11 -103.085 6.2 -23.25; 12 -103.085 6.2 -22.05
13. 13 -102.085 0 -23.25; 14 -102.085 0 -22.05; 15 -102.085 3 -23.25
14. 16 -102.085 3 -22.05; 17 -102.085 6.2 -23.25; 18 -102.085 6.2 -22.05
15. 19 -100.885 0 -23.25; 20 -100.885 0 -22.05; 21 -100.885 3 -23.25
16. 22 -100.885 3 -22.05; 23 -100.885 6.2 -23.25; 24 -100.885 6.2 -22.05
17. MEMBER INCIDENCES
18. 1 5 1; 2 6 2; 3 5 3; 4 6 4; 5 6 5; 6 9 3; 7 7 5; 8 8 6; 9 10 4; 10 9 5
19. 11 10 6; 12 9 7; 13 10 8; 14 10 9; 15 11 9; 16 12 10; 17 12 11; 18 11 17
20. 19 12 18; 20 15 13; 21 16 14; 22 16 15; 23 17 15; 24 18 16; 25 18 17; 26 19 15
21. 27 21 13; 28 20 16; 29 22 14; 30 15 21; 31 16 22; 32 17 23; 33 18 24; 34 21 19
22. 35 22 20; 36 22 21; 37 23 21; 38 24 22; 39 24 23; 40 18 22; 41 24 16; 42 17 21
23. 43 23 15
24. DEFINE MATERIAL START
25. ISOTROPIC STEEL
26. E 2.05E+008
27. POISSON 0.3
28. DENSITY 76.8195
29. ALPHA 1.2E-005
30. DAMP 0.03
31. END DEFINE MATERIAL
32. MEMBER PROPERTY JAPANESE
33. 20 21 23 24 34 35 37 38 TABLE ST H250X250X9
34. 3 4 12 13 TABLE ST H250X250X9
35. 5 10 11 14 22 25 30 TO 33 36 39 TABLE ST H200X200X8
36. 1 2 15 TO 19 TABLE ST C250X90X9
37. 6 TO 9 26 TO 29 40 TO 43 TABLE ST C150X75X6.5
38. CONSTANTS
39. MATERIAL STEEL ALL
40. BETA 180 MEMB 2 16
41. BETA 90 MEMB 3 4 12 13 20 21 23 24 34 35 37 38
42. SUPPORTS
43. 1 TO 4 7 8 13 14 20 PINNED
44. 19 FIXED BUT MY MZ KMX 1E-005
45. MEMBER RELEASE
46. 15 16 END MY MZ
47. 1 2 START MY MZ
48. 17 START MY MZ
49. 17 END MY MZ
50. 33 START MY MZ
51. 33 END MY MZ
52. 32 START MY MZ
53. 32 END MY MZ

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54. 10 START MY MZ
55. 10 END MY MZ
56. 31 END MY MZ
57. 31 START MY MZ
58. 30 START MY MZ
59. 30 END MY MZ
60. 11 START MY MZ
61. 11 END MY MZ
62. MEMBER TENSION
63. 6 TO 9 26 TO 29 40 TO 43
64. LOAD 1 LOADTYPE NONE TITLE DL
65. SELFWEIGHT Y -1 LIST 1 TO 39
66. MEMBER LOAD
67. *0.75KPA DEAD LOAD*
68. 18 19 32 33 UNI GY -0.45
69. 1 2 10 11 15 16 UNI GY -0.375
70. LOAD 2 LOADTYPE NONE TITLE LL
71. *5KPA LIVE LOAD*
72. MEMBER LOAD
73. 18 19 32 33 UNI GY -3
74. 1 2 10 11 15 16 UNI GY -2.5
75. LOAD 3 LOADTYPE NONE TITLE WLX
76. MEMBER LOAD
77. 3 TO 5 12 TO 14 17 20 TO 25 34 TO 39 UNI GX -0.24
78. LOAD 4 LOADTYPE NONE TITLE WLZ
79. MEMBER LOAD
80. 1 TO 4 10 TO 13 15 16 18 TO 21 23 24 30 TO 35 37 38 UNI GZ -0.24
81. 6 TO 9 26 TO 29 UNI GY -0.15
82. LOAD 5 LOADTYPE NONE TITLE NLX
83. SELFWEIGHT X -0.015 LIST 1 TO 39
84. MEMBER LOAD
85. 18 19 32 33 UNI GX -0.007
86. 1 2 10 11 15 16 UNI GX -0.006
87. LOAD 6 LOADTYPE NONE TITLE NLZ
88. SELFWEIGHT Z -0.015 LIST 1 TO 39
89. MEMBER LOAD
90. 18 19 32 33 UNI GZ -0.007
91. 1 2 10 11 15 16 UNI GZ -0.006
92. LOAD 7 LOADTYPE NONE TITLE 1.0DL+1.0LL
93. REPEAT LOAD
94. 1 1.0 2 1.0
95. LOAD 8 LOADTYPE NONE TITLE 1.0DL+1.0LL+1.0WLX
96. REPEAT LOAD
97. 1 1.0 2 1.0 3 1.0
98. LOAD 9 LOADTYPE NONE TITLE 1.0DL+1.0LL+1.0WLZ
99. REPEAT LOAD
100. 1 1.0 2 1.0 4 1.0
101. LOAD 10 LOADTYPE NONE TITLE 1.0DL+1.0LL+1.0NLX
102. REPEAT LOAD
103. 1 1.0 2 1.0 5 1.0
104. LOAD 11 LOADTYPE NONE TITLE 1.0DL+1.0LL+1.0NLZ
105. REPEAT LOAD
106. 1 1.0 2 1.0 6 1.0
107. LOAD 21 LOADTYPE NONE TITLE 1.0DL+1.0LL-1.0 WLX
108. REPEAT LOAD
109. 1 1.0 2 1.0 3 -1.0
110. LOAD 22 LOADTYPE NONE TITLE 1.0DL+1.0LL-1.0 WLZ
111. REPEAT LOAD
112. 4 -1.0 1 1.0 2 1.0
113. LOAD 23 LOADTYPE NONE TITLE 1.0DL+1.0LL-1.0 NLX
114. REPEAT LOAD
115. 1 1.0 2 1.0 5 -1.0
116. LOAD 24 LOADTYPE NONE TITLE 1.0DL+1.0LL-1.0 NLZ
117. REPEAT LOAD
118. 6 -1.0 1 1.0 2 1.0
119. *ULTIMATE LOAD COMBINATIONS
120. LOAD 25 LOADTYPE NONE TITLE 1.0DL+1.6LL
121. REPEAT LOAD
122. 1 1.0 2 1.6
123. LOAD 26 LOADTYPE NONE TITLE 1.4DL+1.6LL

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**START ITERATION NO.          2
**START ITERATION NO.          3
**NOTE-Tension/Compression converged after 3 iterations, Case= 6

**START ITERATION NO.          2
**NOTE-Tension/Compression converged after 2 iterations, Case= 7

**START ITERATION NO.          2
**NOTE-Tension/Compression converged after 2 iterations, Case= 8

**START ITERATION NO.          2
**NOTE-Tension/Compression converged after 2 iterations, Case= 9

**START ITERATION NO.          2
**NOTE-Tension/Compression converged after 2 iterations, Case= 10

**START ITERATION NO.          2
**NOTE-Tension/Compression converged after 2 iterations, Case= 11

**START ITERATION NO.          2
**NOTE-Tension/Compression converged after 2 iterations, Case= 21

**START ITERATION NO.          2
**NOTE-Tension/Compression converged after 2 iterations, Case= 22

**START ITERATION NO.          2
**NOTE-Tension/Compression converged after 2 iterations, Case= 23

**START ITERATION NO.          2
**NOTE-Tension/Compression converged after 2 iterations, Case= 24

**START ITERATION NO.          2
**NOTE-Tension/Compression converged after 2 iterations, Case= 25

**START ITERATION NO.          2

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**START ITERATION NO.      2

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**START ITERATION NO.          2

**NOTE-Tension/Compression converged after 2 iterations, Case= 11
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**START ITERATION NO.          2

**NOTE-Tension/Compression converged after 2 iterations, Case= 21
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```
**START ITERATION NO.          2

**NOTE-Tension/Compression converged after 2 iterations, Case= 22
```

```
**START ITERATION NO.          2

**NOTE-Tension/Compression converged after 2 iterations, Case= 23
```

```
**START ITERATION NO.          2

**NOTE-Tension/Compression converged after 2 iterations, Case= 24
```

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**START ITERATION NO.          2

**NOTE-Tension/Compression converged after 2 iterations, Case= 25

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**START ITERATION NO.      2

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**START ITERATION NO.          2

**NOTE-Tension/Compression converged after 2 iterations, Case= 25

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**START ITERATION NO.      2

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**START ITERATION NO.      2

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**NOTE-Tension/Compression converged after 2 iterations, Case= 26

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 27

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 28

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 29

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 30

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 41

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 42

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 43

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 44

151. DEFINE ENVELOP
152. 25 TO 30 41 TO 44 ENVELOP 1 TYPE STRESS
153. 7 TO 11 21 TO 24 ENVELOP 2 TYPE SERVICEABILITY
154. END DEFINE ENVELOP
155. LOAD LIST 1 TO 11 21 TO 25
156. *PRINT JOINT DISPLACEMENTS ALL
157. LOAD LIST 26 TO 30 41 TO 44
158. *PRINT MEMBER FORCES ALL
159. LOAD LIST ALL
160. PARAMETER 1
161. CODE BS5950
162. SGR 0 MEMB 1 TO 39
163. MAIN 1 MEMB 1 TO 5 10 TO 25 30 TO 39
164. *TRACK 2
165. CHECK CODE MEMB 1 TO 39

STAAD.Pro CODE CHECKING - (BSI)

PROGRAM CODE REVISION V2.12_5950-1_2000

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
=====					
1 ST	C250X90X9	PASS	BS-4.8.3.3.1	0.205	42
		3.07 C	0.60	5.39	-
2 ST	C250X90X9	PASS	BS-4.8.3.3.1	0.206	28
		3.11 C	0.60	5.39	-
3 ST	H250X250X9	PASS	BS-4.3.6	0.014	28
		11.76 C	0.00	3.53	0.00
4 ST	H250X250X9	PASS	BS-4.3.6	0.014	42
		11.18 C	0.00	3.53	0.00
5 ST	H200X200X8	PASS	BS-4.3.6	0.024	42
		0.01 C	0.00	3.53	0.00
6 ST	C150X75X6.5	PASS	BS-4.3.6	0.000	42
		0.00	0.00	0.00	0.00
7 ST	C150X75X6.5	PASS	BS-4.6 (T)	0.003	42
		2.04 T	0.00	0.00	3.16
8 ST	C150X75X6.5	PASS	BS-4.6 (T)	0.004	28
		2.46 T	0.00	0.00	3.16
9 ST	C150X75X6.5	PASS	BS-4.6 (T)	0.001	4
		0.81 T	0.00	0.00	0.00
10 ST	H200X200X8	PASS	BS-4.2.3-(Y)	0.010	26
		6.76 C	0.00	0.00	0.00
11 ST	H200X200X8	PASS	BS-4.2.3-(Y)	0.010	26
		6.76 C	0.00	0.00	0.00
12 ST	H250X250X9	PASS	BS-4.3.6	0.014	28
		22.40 C	0.00	3.65	0.00
13 ST	H250X250X9	PASS	BS-4.3.6	0.014	42
		21.84 C	0.00	3.65	0.00
14 ST	H200X200X8	PASS	ANNEX I.1	0.025	28
		0.01 C	0.00	3.65	-
15 ST	C250X90X9	PASS	BS-4.8.3.3.1	0.248	26
		1.90 C	0.00	8.25	-
16 ST	C250X90X9	PASS	BS-4.8.3.3.1	0.248	26
		1.90 C	0.00	8.25	-
17 ST	C250X90X9	PASS	BS-4.8.3.3.1	0.005	41
		0.01 C	0.05	0.07	-
18 ST	C250X90X9	PASS	BS-4.8.3.3.1	0.131	28
		4.49 C	0.79	5.65	-
19 ST	C250X90X9	PASS	BS-4.8.3.3.1	0.131	42
		4.50 C	0.79	5.65	-
20 ST	H250X250X9	PASS	BS-4.8.2.2	0.046	42
		14.88 T	-1.27	7.85	0.00
21 ST	H250X250X9	PASS	BS-4.8.2.2	0.046	28
		14.28 T	-1.26	-7.85	0.00
22 ST	H200X200X8	PASS	BS-4.8.2.2	0.065	42
		0.02 T	0.00	9.41	0.00

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
=====					
23 ST	H250X250X9	PASS	ANNEX I.1	0.061	28

24	ST	H250X250X9	PASS	ANNEX I.1	0.061	42
		19.99 C		5.65	3.53	-
25	ST	H200X200X8	PASS	ANNEX I.1	0.036	42
		0.03 C		0.79	3.52	-
26	ST	C150X75X6.5	PASS	BS-4.6 (T)	0.008	3
		5.51 T		0.00	0.00	0.00
27	ST	C150X75X6.5	PASS	BS-4.6 (T)	0.034	41
		22.10 T		0.00	0.00	0.00
28	ST	C150X75X6.5	PASS	BS-4.6 (T)	0.008	3
		5.51 T		0.00	0.00	0.00
29	ST	C150X75X6.5	PASS	BS-4.6 (T)	0.034	41
		22.10 T		0.00	0.00	0.00
30	ST	H200X200X8	PASS	BS-4.7 (C)	0.004	26
		7.34 C		0.00	0.00	0.00
31	ST	H200X200X8	PASS	BS-4.7 (C)	0.004	26
		7.34 C		0.00	0.00	0.00
32	ST	H200X200X8	PASS	BS-4.2.3-(Y)	0.014	26
		4.08 C		0.00	0.00	0.00
33	ST	H200X200X8	PASS	BS-4.2.3-(Y)	0.014	26
		4.08 C		0.00	0.00	0.00
34	ST	H250X250X9	PASS	BS-4.7 (C)	0.023	28
		46.91 C		0.00	0.00	3.00
35	ST	H250X250X9	PASS	BS-4.7 (C)	0.023	42
		46.35 C		0.00	0.00	3.00
36	ST	H200X200X8	PASS	BS-4.8.2.2	0.039	28
		0.02 T		0.00	-5.58	1.20
37	ST	H250X250X9	PASS	BS-4.7 (C)	0.009	28
		18.40 C		-0.16	-0.46	3.20
38	ST	H250X250X9	PASS	BS-4.7 (C)	0.009	42
		18.41 C		-0.16	0.46	3.20
39	ST	H200X200X8	PASS	BS-4.2.3-(Y)	0.011	42
		0.03 C		0.00	1.65	0.00

***** END OF TABULATED RESULT OF DESIGN *****

166. FINISH

***** END OF THE STAAD.Pro RUN *****

**** DATE= MAR 25,2013 TIME= 19:32:50 ****

* For questions on STAAD.Pro, please contact *

* Research Engineers Offices at the following locations *

* *

	Telephone	Email
* USA:	+1 (714)974-2500	support@bentley.com
* CANADA	+1 (905)632-4771	detech@odandetech.com
* UK	+44(1454)207-000	support@bentley.com
* NORWAY	+47 67 57 21 30	staad@edr.no
* SINGAPORE	+65 6225-6158	support@bentley.com
* INDIA	+91(033)4006-2021	support@bentley.com
* JAPAN	+81(03)5952-6500	eng-eye@crc.co.jp
* CHINA	+86(411)8479-1166	support@bentley.com
* THAILAND	+66(0)2645-1018/19	support@bentley.com
* North America		support@bentley.com
* Europe		support@bentley.com
* Asia		support@bentley.com
